

[Aim: 100/100 in Maths]

आरंभ

ARITHMETIC PROGRESSION

Lecture #8

ATK :-

A.P.

n^{th} term of AP

$$a_n = a + (n-1)d$$

Sum of ~~first~~ n terms of AP = S_n

$$S_n = \frac{n}{2} (2a + (n-1)d)$$
$$S_n = \frac{n}{2} (a + l)$$

#LP: Find the sum of all natural numbers between 250 and 1000 which are exactly divisible by 3.

div by 3 $\Rightarrow$

① check AP $\checkmark$

250 ————— 1000

252 + 255 + - - - + 999

let 'n' terms

$$\therefore a_n = 999$$

$$a + (n-1)d = 999$$

$$252 + (n-1)(3) = 999$$

$$(n-1)(3) = 999 - 252$$

$$n = 250$$

$$\therefore S_{250} = \frac{n}{2}(a+l)$$

$$= \frac{250}{2}(252 + 999)$$

$$= 125 \times 1251$$

$$S_{250} \Rightarrow 125 \times 1251$$

$$\Rightarrow \underline{\hspace{2cm}}$$

#LP: If S_n is the sum of first n terms of an A.P., is given by $S_n = 5n^2 + 3n$, then find its n^{th} term. [CBSE 2009]

$$S_n = 5n^2 + 3n$$

$$n=1 \rightarrow S_1 = 5(1)^2 + 3(1)$$

$$S_1 = 8 = a_1 = a = 8$$

$$n=2 \rightarrow S_2 = 5(2)^2 + 3(2)$$

$$S_2 = 20 + 6$$

$$a_1 + a_2 = 26$$

$$8 + a_2 = 26$$

$$a_2 = 18$$

$$\therefore d = a_2 - a_1$$

$$= 18 - 8$$

$$d = 10$$

To find

$$a_n = a + (n-1)d$$

$$a_n = 8 + (n-1)(10)$$

$$a_n = 8 + 10n - 10$$

$$a_n = 10n - 2$$



#LP: If the m^{th} term of an A.P. is 1 and the n^{th} term is 1, show that

the sum of mn terms is $\frac{1}{2}(mn+1)$

$$a_m = \frac{1}{n} \Rightarrow a + (m-1)d = \frac{1}{n} \Rightarrow a + md - d = \frac{1}{n}$$

$$a_n = \frac{1}{m} \Rightarrow a + (n-1)d = \frac{1}{m} \Rightarrow a + nd - d = \frac{1}{m}$$

$$md - nd = \frac{1}{n} - \frac{1}{m}$$

$$d(m-n) = \frac{m-n}{mn}$$

$$d = \frac{1}{mn}$$

$$\rightarrow a + (n-1)\left(\frac{1}{mn}\right) = \frac{1}{m}$$

$$\rightarrow a + n \times \frac{1}{mn} - \frac{1}{mn} = \frac{1}{m}$$

$$\rightarrow a + \frac{1}{n} - \frac{1}{mn} = \frac{1}{m}$$

$$\rightarrow a = \frac{1}{mn}$$

$$S_{mn} = \frac{1}{2}(mn+1)$$

$$\text{LHS } S_{mn} = \frac{mn}{2} (2a + (mn-1)d)$$

$$S_{mn} = \frac{mn}{2} \left(2\left(\frac{1}{mn}\right) + (mn-1)\left(\frac{1}{mn}\right) \right)$$

$$S_{mn} = \frac{mn}{2} \left[\frac{2}{mn} + \frac{1}{mn} - \frac{1}{mn} \right]$$

$$S_{mn} = \frac{mn}{2} \left[\frac{2}{mn} - \frac{1}{mn} + \frac{1}{mn} \right]$$

$$S_{mn} = \frac{mn}{2} \left[\frac{2-1+1}{mn} \right]$$

$$S_{mn} = \frac{mn}{2} \left[\frac{1+mn}{mn} \right] \Rightarrow \frac{1}{2}(mn+1) \text{ (HP)}$$

#LP : The sum of $n, 2n, 3n$ terms of an A.P. are S_1, S_2, S_3 respectively. Prove that :

$$S_3 = 3(S_2 - S_1)$$

$$S_n = S_1$$

$$\frac{n}{2} [2a + (n-1)d] = S_1$$

$$S_{2n} = S_2$$

$$\frac{2n}{2} [2a + (2n-1)d] = S_2$$

$$S_{3n} = S_3$$

$$\frac{3n}{2} [2a + (3n-1)d] = S_3$$

To prove

$$S_3 = 3(S_2 - S_1)$$

RHS

$$3(S_2 - S_1)$$

$$\Rightarrow 3 \left[\frac{2n}{2} [2a + (2n-1)d] - \frac{n}{2} [2a + (n-1)d] \right]$$

$$\Rightarrow 3 \left[\frac{n}{2} [2(2a + 2nd - d) - (2a + nd - d)] \right]$$

$$\Rightarrow \frac{3n}{2} [4a + 4nd - 2d - 2a - nd + d]$$

$$\Rightarrow \frac{3n}{2} [2a + 3nd - d]$$

$$\Rightarrow \frac{3n}{2} [2a + (3n-1)d] = S_3 \text{ (HP)}$$

#LP: If the sum of m terms of an A.P. is the same as the sum of its n terms, show that the sum of its $(m+n)$ terms is zero. [CBSE 2017]

given for same AP, $\begin{matrix} \nearrow a \\ \searrow d \end{matrix}$

$$S_m = S_n$$

$$\Rightarrow \frac{m}{2} [2a + (m-1)d] = \frac{n}{2} [2a + (n-1)d]$$

$$\Rightarrow m[2a + md - d] = n[2a + nd - d]$$

$$\Rightarrow \underline{2am} + \underline{m^2d} - \underline{md} = \underline{2an} + \underline{n^2d} - \underline{nd}$$

$$\Rightarrow \underline{2am} - \underline{2an} = \underline{n^2d - m^2d} + \underline{md} - \underline{nd}$$

$$\Rightarrow 2a(m-n) = d(n^2 - m^2) + d(m-n)$$

$$\Rightarrow 2a(m-n) = d \underline{(n-m)(n+m)} + d \underline{(m-n)}$$

$$+ 2a(m-n) = \underline{-d(m-n)(n+m)} + d \underline{(m-n)}$$

$$\Rightarrow \underline{2a(m-n)} = \underline{(m-n)} [-d(n+m) + d]$$

$$\Rightarrow 2a = d(-(n+m) + 1)$$

$$\Rightarrow 2a = d(-n-m+1)$$

$$\Rightarrow \underline{2a = -d(m+n-1)}$$

To prove

$$S_{m+n} = 0$$

$$\frac{m+n}{2} [2a + (m+n-1)d] = 0$$

$$\text{LHS} \quad \frac{m+n}{2} \left[\underline{-d(m+n-1)} + \underline{(m+n-1)d} \right]$$

$$\Rightarrow 0 \quad \text{RHS}$$

#LP: The ratio of the sums of m and n terms of an A.P. is $m^2 : n^2$.

Show that the ratio of the m^{th} and n^{th} term is $(2m - 1) : (2n - 1)$.

[CBSE 2016, 2017]

$$\Rightarrow \frac{S_m}{S_n} = \frac{m^2}{n^2}$$

$$\Rightarrow \frac{\frac{m}{2} [2a + (m-1)d]}{\frac{n}{2} [2a + (n-1)d]} = \frac{m^2}{n^2}$$

$$\Rightarrow \frac{2a + md - d}{2a + nd - d} = \frac{m}{n}$$

$$\Rightarrow 2an + md - nd = 2am + nd - md$$

$$\Rightarrow 2an - 2am = nd - md$$

$$\Rightarrow 2a(n-m) = d(n-m)$$

$$\Rightarrow \boxed{2a = d} \text{ or } \boxed{d = 2a}$$

To prove

$$\frac{a_m}{a_n} = \frac{2m-1}{2n-1}$$

LHS $\Rightarrow \frac{a + (m-1)d}{a + (n-1)d} \Rightarrow \frac{a + (m-1)(2a)}{a + (n-1)(2a)}$

$$\Rightarrow \frac{a[1 + (m-1)2]}{a[1 + (n-1)2]}$$

$$\Rightarrow \frac{1 + 2m - 2}{1 + 2n - 2} \Rightarrow \frac{2m-1}{2n-1}$$

HP

#LP: The sum of the first 7 terms of an A.P is 63 and the sum of its next 7 terms is 161. Find the 28th term of this AP. [CBSE 2014]

S_{14} ✓
 $a_1, a_2, a_3, a_4, a_5, a_6, a_7, a_8, a_9, a_{10}, a_{11}, a_{12}, a_{13}, a_{14}, \dots$

$S_7 = 63$ ★ ★ ★ ★ ★ ★ ★ 161

Formula apply

$S_7 = 63$ ✓

$\Rightarrow \frac{7}{2} [2a + (7-1)d] = 63$

$\Rightarrow 2a + 6d = 18$

$\cancel{2(a+3d)} = 18$

$a + 3d = 9$ (I)

~~$S_7 = 161$~~

$S_8 \rightarrow 14 = 161$

$S_{14} - S_7 = 161$

$\frac{14}{2} [2a + (14-1)d] - \frac{7}{2} [2a + (7-1)d] = 161$

$\frac{7}{2} [2(2a+13d) - (2a+6d)] = 161$

$\rightarrow 4a + 26d - 2a - 6d = 23 \times 2$

$\rightarrow 2a + 20d = 23 \times 2$

$\cancel{2(a+10d)} = 23 \times 2$

$\rightarrow a + 10d = 23$ (II)

$$a + 10d = 23$$

$$a + 3d = 9$$

$$7d = 14$$

put

$$d = 2$$



$$a + 3(2) = 9$$

$$a + 6 = 9$$

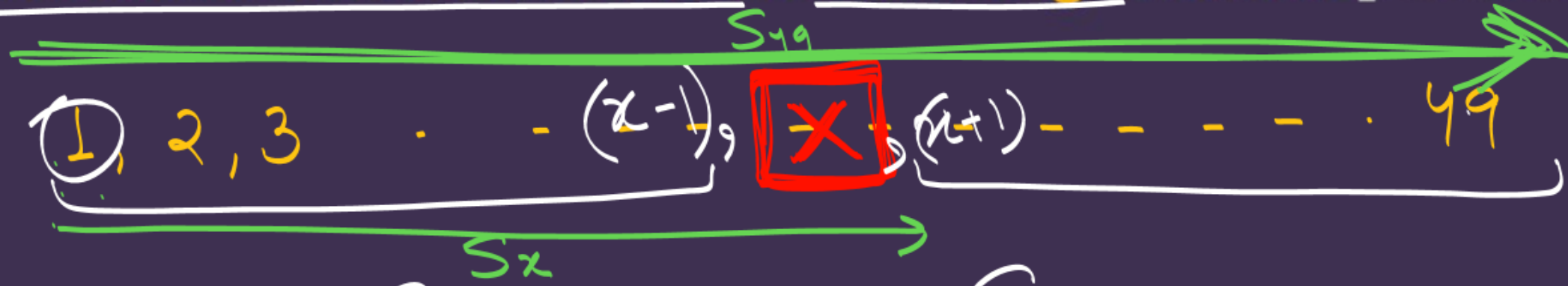
$$a = 3$$

Final ans

$$\begin{aligned} a_{28} &= a + 27d \\ &= (3) + 27(2) \\ &= 3 + 54 \end{aligned}$$

$$a_{28} = 57$$

#LP: The houses in a row are numbered consecutively from 1 to 49. There exists a house x such that the sum of numbers of houses preceding the house numbered x is equal to sum of the numbers of houses following x . Find x . [CBSE 2016]



Direct

S_n formula ✓

$$S_1 \rightarrow (x-1) = S_{(x+1) \rightarrow 49}$$

$$S_{(x-1)} = S_{49} - S_x$$

$$\Rightarrow \frac{(x-1)}{2} [2a + (x-1-1)d] = \frac{49}{2} [2a + (49-1)d] - \left[\frac{x}{2} (2a + (x-1)d) \right]$$

$$\Rightarrow \frac{(x-1)}{2} [2(1) + (x-2)(1)] = \frac{1}{2} [49(2(1) + 48(1))] - [x(2(1) + (x-1)(1))]$$

$$\Rightarrow (x-1) \left(\frac{2}{2} + x - \frac{2}{2} \right) = \left[\frac{49(50)}{2} - x(1+x) \right]$$

$$\Rightarrow (x-1)(x) = [49 \times 50 - x(1+x)]$$

$$\begin{matrix} a=1 \\ d=1 \end{matrix}$$

$$\rightarrow x(x-1) = 2450 - x(x+1)$$

$$\rightarrow x^2 - \cancel{x} = 2450 - x^2 - \cancel{x}$$

$$\rightarrow x^2 + x^2 = 2450$$

$$\rightarrow \cancel{2}x^2 = \cancel{2450} \quad | \quad 225$$

$$\rightarrow x^2 = 1225$$

$$\rightarrow x = \sqrt{1225}$$

$$x = \pm 35$$

$$x = +35$$

∴ 35th house

$x = -35$ / house no. cannot be -ve

THANK YOU COODIESS !!

